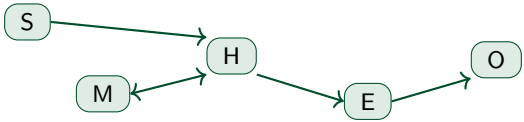
Great Lakes



S = Superior M = Michigan H = Huron E = Erie O = Ontario

Lake	Volume $10^{12}\,\mathrm{m}^3$	Precip. $10^3\,\mathrm{m}^3/\mathrm{s}$	Watershed $10^3\,\mathrm{m}^3/\mathrm{s}$	Evap. $10^3\,\mathrm{m}^3/\mathrm{s}$	Lake inflow $10^3\,\mathrm{m}^3/\mathrm{s}$	Lake outflow $10^3\,\mathrm{m}^3/\mathrm{s}$	Other outflow $10^3\,\mathrm{m}^3/\mathrm{s}$
S	12.10	+2.00	+1.50	−1.40	—	−2.10 H	—
M	4.92	+1.50	+1.29	−1.10	+6.00 H +2.10 S	−7.60 H −6.00 M	−0.09 Chicago
H	3.54	+1.60	+1.40	−1.10	+7.60 M	−5.60 E	—
E	0.484	+0.70	+0.70	−0.70	+5.60 H	−6.30 O	—
O	1.64	+0.50	+1.00	−0.40	+6.30 E	—	−7.40 St. Lawrence

Volume balance

With t in seconds and each $v_i(t)$ in m^3

$$
\begin{aligned}
v_s'(t) &= (+2.00 + 1.50 - 1.40 - 2.10) \times 10^3 = 0 \\
v_m'(t) &= (+1.50 + 1.29 - 1.10 + 6.00 - 7.60 - 0.09) \times 10^3 = 0 \\
v_h'(t) &= (+1.60 + 1.40 - 1.10 + 2.10 + 7.60 - 6.00 - 5.60) \times 10^3 = 0 \\
v_e'(t) &= (+0.70 + 0.70 - 0.70 + 5.60 - 6.30) \times 10^3 = 0 \\
v_o'(t) &= (+0.50 + 1.00 - 0.40 + 6.30 - 7.40) \times 10^3 = 0
\end{aligned}
$$

Lake volumes are constant.

$$
\begin{aligned}
v_s &= 12.10 \times 10^{12}\,\mathrm{m}^3 \qquad v_m = 4.92 \times 10^{12}\,\mathrm{m}^3 \qquad v_h = 3.54 \times 10^{12}\,\mathrm{m}^3 \\
v_e &= 0.484 \times 10^{12}\,\mathrm{m}^3 \qquad v_o = 1.64 \times 10^{12}\,\mathrm{m}^3
\end{aligned}
$$

Pollutant balance in well-mixed lakes

- Lakes are well-mixed tanks with constant volume v in m^3.
- Pollutants do not evaporate and change only through water transfers, no external supply.
- Pollutants $s(t)$ in kg and concentration $c(t) = s(t)/v$ in $\mathrm{kg/m^3}$. Initial concentration c_0 gives $s(0) = vc_0$.
- An inflow q_i in $\mathrm{m^3/s}$ carries the source concentration c_i. Outflow q_{out} (excluding evaporation) carries the lake's concentration $s(t)/v$.

Flow times concentration gives pollutant rate $(\cancel{\mathrm{m}^3}/\mathrm{s})(\mathrm{kg}/\cancel{\mathrm{m}^3}) = \mathrm{kg/s}$.

$$s'(t) = \underbrace{\sum_{\text{incoming flows}} q_i c_i}_{\text{pollutant inflow rate}} - \underbrace{q_{\mathrm{out}} s(t)/v}_{\text{pollutant outflow rate}}$$

Five-Lake matrix A

With pollutants $\boldsymbol{s} = (s_s, s_m, s_h, s_e, s_o)$ in kg and time t in seconds. Lake Michigan receives pollution from Lake Huron and loses it to Lake Huron and the Chicago river. The ODE is

$$s_m' = 10^{-9}\big(6.00 s_h/3.54 - (7.60 + 0.09) s_m/4.92\big)$$

Each row is pollution balance for the matching lake.

$$\boxed{\boldsymbol{s}' = A\boldsymbol{s}} \qquad A = 10^{-9}\begin{pmatrix} -2.10/12.10 & 0 & 0 & 0 & 0 \\ 0 & -7.69/4.92 & 6.00/3.54 & 0 & 0 \\ 2.10/12.10 & 7.60/4.92 & -11.60/3.54 & 0 & 0 \\ 0 & 0 & 5.60/3.54 & -6.30/0.484 & 0 \\ 0 & 0 & 0 & 6.30/0.484 & -7.40/1.64 \end{pmatrix}$$

Each entry in A has units s^{-1} since

$$\frac{10^3\,\mathrm{m}^3/\mathrm{s}}{10^{12}\,\mathrm{m}^3} = 10^{-9}\,\mathrm{s}^{-1}$$

Eigenvalues and eigenvectors of A

Eigenvalues are

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5) = 10^{-9}(-0.17355, -0.58895, -4.25090, -4.51220, -13.01653)$$

with units s^{-1}. The eigenvectors are columns of

$$W = [\boldsymbol{w}_1\ \boldsymbol{w}_2\ \boldsymbol{w}_3\ \boldsymbol{w}_4\ \boldsymbol{w}_5] = \begin{pmatrix} 0.9743 & 0 & 0 & 0 & 0 \\ 0.1692 & 0.8468 & -0.0695 & 0 & 0 \\ 0.1387 & 0.4867 & 0.1103 & 0 & 0 \\ 0.0171 & 0.0619 & 0.0199 & 0 & -0.5470 \\ 0.0513 & 0.2055 & 0.9913 & 1.0000 & 0.8372 \end{pmatrix}$$

with rows are ordered by lake s, m, h, e, o. The columns satisfy

$$A\boldsymbol{w}_k = \lambda_k \boldsymbol{w}_k \qquad AW = W\,\mathrm{diag}(\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5)$$

General solution and long-time concentrations

The general solution is

$$\boldsymbol{s}(t) = \sum_{k=1}^{5} a_k e^{\lambda_k t} \boldsymbol{w}_k \qquad \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{pmatrix} = W^{-1}\boldsymbol{s}(0)$$

Provided $a_1 \neq 0$, the slowest-decaying mode dominates as $t \to \infty$

$$\boldsymbol{s}(t) = a_1 e^{\lambda_1 t}\boldsymbol{w}_1 + O(e^{\lambda_2 t}) = a_1 e^{-0.17355\times 10^{-9} t}\boldsymbol{w}_1 + O(e^{\lambda_2 t})$$

All concentrations approach zero. Asymptotically in the slowest mode concentration ratios

$$\left(\frac{c_s(t)}{c_s(t)}, \frac{c_m(t)}{c_s(t)}, \frac{c_h(t)}{c_s(t)}, \frac{c_e(t)}{c_s(t)}, \frac{c_o(t)}{c_s(t)}\right) \longrightarrow (1,\ 0.427,\ 0.487,\ 0.438,\ 0.388)$$

Michigan's concentration is $\sim 43\%$ of Superior's, Huron's 49%, Erie's 44%, and Ontario's 39%

Estimation notes

- Volumes are rounded and balanced estimates from 1953–2010 data.
- Flows are rounded and adjusted to balance inflow and outflow.
- Michigan–Huron precipitation/evaporation are split by surface/watershed area.
- Assumed Mackinac flow split (m^3/s) is 7600 to Huron and 6000 to Michigan.
- Superior watershed includes Long Lac and Ogoki.
- Huron to Erie ignores tiny Lake St. Clair. Erie to Ontario combines the Niagara River and Welland Canal.

Mackinac exchange estimate

The initial directional estimates used an assumed oscillating transport

$$Q(t) = 1100 + 20000 \sin(2\pi t/T) \qquad [Q] = \mathrm{m}^3/\mathrm{s}$$

Positive transport is toward Huron. Averaging over a complete cycle gives

$$\langle \max(Q(t), 0) \rangle \approx 6930 \ \mathrm{m}^3/\mathrm{s}$$
$$\langle \max(-Q(t), 0) \rangle \approx 5830 \ \mathrm{m}^3/\mathrm{s}$$

The final teaching values were adjusted to 7600 and 6000 m^3/s to balance the five-lake model. These are assumed directional averages, not measured long-term averages.

References

1. University of Michigan, Graham Sustainability Institute (2015).
   Great Lakes Water Budget, Figure 1.
   Historical water-budget components and connecting flows from NOAA-GLERL data.
2. U.S. Environmental Protection Agency.
   Physical Features of the Great Lakes.
   Surface and land drainage areas for the initial Michigan–Huron split.
3. NOAA Technical Memorandum GLERL-137, Table 3.
   Modified Great Lakes Hydrology Modeling System for Considering Simple Extreme Climates.
   Coordinated lake volumes, including Georgian Bay within Huron.

References for Mackinac exchange

4. Anderson and Schwab (2013).
   Predicting the oscillating bi-directional exchange flow in the Straits of Mackinac.
   Net transport near 1100 m^3/s and reversing exchange.
5. U.S. Environmental Protection Agency.
   Biological, Chemical and Physical Relationships in the Straits of Mackinac.
   Observed transport magnitudes motivating the assumed oscillation amplitude.
6. Anderson and Schwab (2017).
   Meteorological influence on summertime baroclinic exchange in the Straits of Mackinac.
   Simultaneous opposing flows during summer stratification.